

Nagizaaz Shaik

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PROFESSIONAL SUMMARY

AI / ML Engineer with 5+ years in data and ML engineering, including 2+ years building production LLM systems, multi-agent pipelines, and secure ML platforms at scale. Deep specialization in RAG architectures, agentic orchestration (LangGraph, CrewAI), LLMops, and AI safety — with a track record of shipping inference infrastructure, evaluation frameworks, and generative-AI products in regulated financial environments. AWS Machine Learning Specialty certified.

EDUCATION

Belhaven University

Doctorate of Business Administration (DBA), Applied Artificial Intelligence Specialization.

Jackson, MS

Aug 2026 – Present

Relevant Coursework: Applied Artificial Intelligence, Machine Learning for Decision-Making, AI Strategy & Governance, Data Analytics, Business Research Methods

University of Dayton

Master of Science, Computer Science. GPA: 3.93 / 4.0.

Dayton, OH

Aug 2023 – May 2025

Relevant Coursework: Advanced Machine Learning, NLP, Human-AI Interaction, Big Data Systems, Cloud Computing, Distributed Systems, Statistical Inference

CMR Engineering College

Bachelor of Technology, Computer Science Engineering. GPA: 8.9 / 10.

Hyderabad, India

Aug 2016 – May 2020

Relevant Coursework: Data Structures & Algorithms, DBMS, OOP (Java), Computer Networks, Operating Systems, Software Engineering, Design & Analysis of Algorithms

EXPERIENCE

AI/ML Platform Architect

Jul 2025 – Jun 2026

Wright-Patt Credit Union

Fairborn, OH

- Designed the organization's AWS ML platform end to end: SageMaker inference endpoints, MLflow model registry with version-gated promotion and automated rollback, and per-environment VPC isolation, reducing deployment failures 85% within 60 days.
- Defined data-security architecture for all ML workloads: AES-256 at rest and in transit, CloudTrail on every inference call, IAM scoped to least privilege; passed two consecutive NCUA audits with zero findings on \$400M in member financial data.
- Introduced an internal LLM gateway supporting Claude, OpenAI, and Ollama with evaluation guardrails, cost controls, and audit logging; RAGAS-based evaluation showed a 22% lift in grounded-answer accuracy over the prior single-model baseline, enabling application teams to adopt generative-AI features against approved models.
- Codified infrastructure in Terraform and migrated deployments to a GitOps workflow via ArgoCD, enabling reproducible, peer-reviewed environment changes and faster, safer releases.
- Mentored two engineers into ML and platform ownership through weekly architecture and code-review sessions; both now ship production components independently.

ML Engineer

Jul 2024 – Jun 2025

Day Air Credit Union

Kettering, OH

- Built and deployed an XGBoost credit-risk model on AWS SageMaker scoring 500+ consumer-loan applications per day, automating first-pass underwriting and cutting manual analyst review 30% while holding approval-rate stability within risk thresholds.
- Fine-tuned a BERT classifier on member support tickets and shipped it as a FastAPI microservice: 14 intent categories, sub-200ms p99 latency, routing 10,000+ weekly requests with a 40% drop in misrouting.
- Built a RAG document-QA system over 500+ internal policy and NCUA regulatory documents using LangChain, vector search, and Claude 3.5, reducing compliance research time from 2 hours to under 20 minutes per query.
- Engineered a weekly automated retraining pipeline in Apache Airflow with rolling-window retrain, holdout evaluation, and promotion gates on AUC, precision, and calibration, replacing 8+ hours of manual work per cycle.
- Deployed model monitoring via SageMaker Model Monitor with custom statistical baselines, catching two data drift events before they affected production credit decisions.

Senior Data Engineer

Jul 2020 – Jun 2023

Cognizant

Hyderabad, India

- Led an Oracle 11g to Amazon Redshift migration for a financial-services client: redesigned the schema for columnar storage with tuned distribution and sort keys and re-platformed 20+ stored procedures as modular Airflow DAGs, cutting average analytical query runtime 60% across a 25 GB+ warehouse serving five business units.
- Designed 15 production ETL/ELT pipelines in Apache Airflow across Oracle, SQL Server, flat files, and REST APIs, with a multi-stage validation layer at every hop; surfaced a silent upstream corruption that had been invalidating three months of downstream financial reports.
- Built a real-time Kafka feature pipeline aggregating POS events in five-minute tumbling windows and landing to Redshift at sub-60-second end-to-end latency, contributing to a 28% reduction in emergency restocking.
- Promoted from Associate to Senior Data Engineer within two years; led code reviews for three engineers, conducted weekly data-modelling sessions, and served as primary client contact across two accounts with 100% on-time delivery.

- Built SQL queries and Vertica views powering weekly driver-utilisation, trip-completion, and surge-pricing reporting; standardised ad-hoc queries into reusable templates, cutting analyst prep time from 3 hours to under 45 minutes.
- Automated extraction and transformation of operational event data with Python and pandas, turning a 4-hour manual weekly process into a scheduled job; built an anomaly-detection script flagging unusual trip counts against rolling 7-day baselines, earning a return internship offer.

PROJECTS**LLM Gateway — github.com/shaikn6/llm-gateway**

- Production LLM gateway with an OpenAI-compatible API: semantic caching, latency-aware routing across Claude / OpenAI / Ollama, token-budget enforcement, and per-key cost analytics behind a single FastAPI interface with Prometheus observability.
- *Stack: FastAPI, Claude, OpenAI, Ollama, Redis, Prometheus, Docker*

Finance Agent Crew — github.com/shaikn6/finance-agent-crew

- Multi-agent financial research system where parallel analyst, risk-assessor, and report-writer agents collaborate over SEC filings, earnings calls, and news to produce structured investment briefs with citation chains, orchestrated in LangGraph with n8n automation.
- *Stack: LangGraph, Claude, SEC EDGAR API, FAISS, n8n, FastAPI*

LLM Safety Auditor — github.com/shaikn6/llm-safety-auditor

- Automated red-teaming framework for production LLMs with 250+ adversarial attack vectors across six mutation strategies (jailbreak, indirect prompt injection, encoding obfuscation, multi-turn escalation); scores against the full OWASP LLM Top 10 and generates compliance-grade PDF audit reports.
- *Stack: Python, HuggingFace, OWASP LLM Top 10, Red-Teaming, FastAPI, ReportLab*

nano-finbert — github.com/shaikn6/nano-finbert

- Tiny transformer encoder (~2M params) trained from scratch on financial text with no pretrained weights and no HuggingFace dependency; outputs structured market signals: sentiment, entities, sectors, event type, and impact score.
- *Stack: PyTorch, Transformers, NLP, Fintech*

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty, Amazon Web Services
- AWS Certified Cloud Practitioner, Amazon Web Services
- GCP Professional Data Engineer, Google Cloud
- Snowflake SnowPro Advanced: Data Engineer, Snowflake
- dbt Analytics Engineering Certification, dbt Labs

SKILLS

LLM & Agentic AI: RAG (Hybrid Search, Reranking, HyDE, Corrective RAG), Multi-Agent Orchestration (LangGraph, CrewAI), Model Context Protocol (MCP), Prompt Engineering & Versioning, Semantic Caching, LLM Gateways, OWASP LLM Top 10, Red-Teaming

LLMOps & Eval: RAGAS, LLM-as-Judge, A/B Testing, Cost Analytics, Agent Observability, MLflow, NVIDIA NIM, vLLM / Ollama

ML & AI: PyTorch, Transformers (from-scratch), XGBoost, BERT, HuggingFace, scikit-learn, SHAP, AWS SageMaker (Pipelines, Model Monitor)

Cloud & Infra: AWS (SageMaker, EC2, S3, RDS, Glue, Redshift, EKS, IAM, CloudWatch, CloudTrail), GCP (BigQuery, Vertex AI), Terraform

MLOps: Apache Airflow, MLflow Registry, SageMaker Pipelines, Docker, Kubernetes, ArgoCD, CI/CD, FastAPI

Data Governance & Security: NCUA / SOC 2-aware, ECOA / Fair-Lending testing, PII handling (Presidio), AES-256 (rest/transit), OPA/Conftest, Checkov, Gitleaks, SLSA

Data Engineering: Kafka, Spark / PySpark, dbt, AWS Glue, Redshift, Snowflake, Oracle SQL / PL-SQL

Languages: Python, SQL, PL/SQL, Bash